SPM Module 2: Partitioned Hash Join with Duplicates

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1 Parallelization Strategy

The partitioned hash join pipeline consists of five phases: *Histogram*, *Prefix Sum*, *Scatter*, *Join Local*, and a final sequential *Accumulation*. Each phase has distinct computational and memory characteristics that determine whether parallelism is beneficial and which strategy to adopt.

Thread Team and Barrier Synchronization

A single team of k threads is launched once and reused across all phases via `std::barrier` (C++20). An alternative design would use a thread pool with a task queue, but it would introduce unnecessary indirection: because all threads must finish each phase before the next one can begin — due to data dependencies between histogram, prefix sum, scatter, and join — a full barrier is required at every phase boundary regardless. The barrier approach directly models this bulk-synchronous structure and avoids the overhead of thread pool bookkeeping (task submission, queue polling, wakeup latency). Spawning and joining threads at each phase boundary would inflate the sequential component f in Amdahl's model by $O(k)$ spawn/join cycles per phase. The barrier completion function runs a single delegate thread between phases to perform the sequential prefix sum and to compute per-thread scatter offsets from the local histograms.

Adaptive thread count. To prevent synchronisation overhead from dominating small workloads, the actual thread count k is:

$$k = \min(k_{\max},\ \lfloor \min(NR, NS) \,/\, T_{\min} \rfloor)$$

where $k_{\max}$ is the user-requested count and $T_{\min} = 8\,192$ records is the minimum per-thread work threshold. Below this threshold, the cost of barrier synchronisation and cache-line traffic between threads exceeds the parallelism gain: with $N < k \cdot T_{\min}$, each thread's scan completes in less time than the $O(k \cdot P)$ histogram-merge step executed by the barrier delegate, so synchronisation overhead dominates useful work. This threshold is never reached in the benchmarks below ($NR \geq 10^6$), so all experiments run with the full requested thread count.

Histogram (Phases 0 and 2)

Each thread processes a contiguous block of $\lceil N/k \rceil$ records from relation R (or S), incrementing its own private histogram of size P. Thread-local histograms eliminate all write contention. After the barrier, the completion delegate merges the k local histograms into a global one in $O(P \cdot k)$ time, then runs the prefix sum $O(P)$ and computes per-thread scatter offsets.

Static block distribution yields good spatial locality: each thread reads a contiguous slice of the input array, maximising cache line utilisation. The dominant cost is a sequential scan of $N \times 8\,\mathrm{B}$ — memory-bandwidth-bound ($I \approx 0.125\,\mathrm{FLOP/byte}$), so the histogram phase scales poorly (Section 3). The prefix sum over the P-element global histogram is performed sequentially by the barrier delegate ($< 0.1\%$ of total time at $P \leq 1024$; parallelising it would not amortise synchronisation overhead).

Scatter (Phases 1 and 3)

Scatter rearranges records into partition-contiguous output arrays. Lock-free parallelism is possible because per-thread scatter offsets are fully determined during the barrier: thread t starts writing partition p at offset

$$\mathrm{off}[t][p] = \mathrm{global\_begin}[p] + \sum_{j=0}^{t-1} \mathrm{local\_hist}[j][p].$$

Each thread thus writes to a non-overlapping, predetermined region of the output array with no synchronisation needed at runtime:

```
auto cursor = offsets_R[t]; // local copy on stack
for (size_t i = r_start; i < r_end; ++i) {
    uint32_t pid = hash_key(R[i].key, shift);
    out_R[cursor[pid]++] = R[i];
}
```

Scatter is memory-bound (random writes driven by hash values), but the absence of any contention allows good parallel throughput. Partition assignment uses the Fibonacci multiply-shift hash from Module 1 (`hash_key`: $h(k) = ((k_{\mathrm{lo}} \oplus k_{\mathrm{hi}}) \cdot A_{32}) >> s$, $A_{32} = \lfloor 2^{32}/\varphi \rfloor$) rather than the reference $k \bmod P$, giving uniform partition sizes even for structured key sets.

Join Local (Phase 4)

Each partition can be joined independently: a flat open-addressing hash table `FlatCountMap` is built on the R slice, then the S slice is probed. This is an embarrassingly parallel workload. At the throughput-optimal $P = 512$, each per-partition `FlatCountMap` occupies $\approx 1\,\mathrm{MB}$ and is processed one at a time per thread; the table fits within the 20 MB aggregate shared L3, making this phase compute-bound (hash arithmetic and cache-resident lookups) rather than DRAM-bound. Partitions are distributed cyclically: thread t owns partitions t, $t{+}k$, $t{+}2k$, …, requiring neither scheduling logic nor atomics.

```
for (uint32_t pid = t; pid < P; pid += nt) {
    size_t rb = global_begin_R[pid], re = rb + global_hist_R[
        pid];
    size_t sb = global_begin_S[pid], se = sb + global_hist_S[
        pid];
    if (rb == re || sb == se) continue;
    FlatCountMap cntR(re - rb); // build on R slice
    for (size_t i = rb; i < re; ++i) cntR.increment(out_R[i].
        key);
    for (size_t i = sb; i < se; ++i) // probe with S
        local.join_count += cntR.count(out_S[i].key);
    // checksum fields omitted for clarity
}
```

For the Fibonacci partitioning hash, partition sizes are statistically near-equal, so cyclic assignment achieves the same load balance as LPT scheduling — with zero scheduling overhead. LPT would require sorting all P partition sizes ($O(P \log P)$) before distribution; since sizes are near-uniform, that cost yields no benefit. Per-thread result accumulators are padded to 64 bytes (`alignas(64)`) to prevent false sharing. After the barrier, the main thread reduces the partial results.

2 Correctness Verification

Correctness is verified at two levels. For small inputs ($NR \leq 500$, $NS \leq 500$), both the sequential and parallel binaries automatically run a naive $O(N^2)$ reference join and compare `join_count`, `checksum1`, and `checksum2`:

```
./hashjoin_par -nr 50 -ns 80 -seed 13 \
  -max-key 8 -p 4 -t 4
# output: naive_verify=PASS
```

For large inputs, the same three output values produced by the sequential and parallel binaries are compared across all tested thread counts. Since `checksum1` and `checksum2` use independent hash multipliers, the probability of a false positive is negligible. The verification is outside the measured region and does not affect reported timings.

3 Performance Results

All experiments were run on a node of the `spmcluster`: Intel Xeon E5-2640 v2 @ 2.00 GHz, 2 sockets × 8 physical cores (16 physical, 32 with Hyper-Threading), NUMA architecture. Default parameters: `seed=42`, $NS = 2 \times NR$, `max_key=1M`, `P=128`; best of 5 repetitions, no CPU pinning (run-to-run variation $\leq 3\%$). Strong and weak scaling fix $P = 128$ to isolate the effect of thread count from partition count; the partition sensitivity analysis (below) independently characterises the effect of P and identifies $P = 512$ as the throughput-optimal value. Thread counts $p \in \{1, 2, 4, 8, 12, 16, 20, 22, 24, 26, 28, 30, 32\}$ cover the physical-core regime ($p \leq 16$) and the Hyper-Threading regime ($p > 16$) with gaps of at most $\Delta p = 4$ between consecutive points; the phase breakdown additionally includes $p = 14$ to sample performance just below the 16-physical-core limit.

Strong Scaling

p	NR=10M		NR=20M	
	$S(p)$	$E(p)$	$S(p)$	$E(p)$
seq	1.00	100%	1.00	100%
1	1.03	103%	1.04	104%
2	1.40	70%	1.45	72%
4	2.61	65%	2.60	65%
8	4.46	56%	4.49	56%
12	6.04	50%	6.24	52%
16	7.49	47%	7.59	47%
20	7.41	37%	7.33	37%
22	7.51	34%	8.27	38%
24	7.93	33%	8.65	36%
26	8.34	32%	9.15	35%
28	8.50	30%	9.32	33%
30	8.38	28%	9.32	31%
32	8.64	27%	9.64	30%

Shaded: HT regime ($p > 16$). $S(1) > 1.0$: seq/par are distinct executables; best-of-5 minima diverge by $\leq 4\%$ systematically (within-run variance $\leq 3\%$). The dip at $p = 20$ is a NUMA effect (Section 4).

Weak Scaling

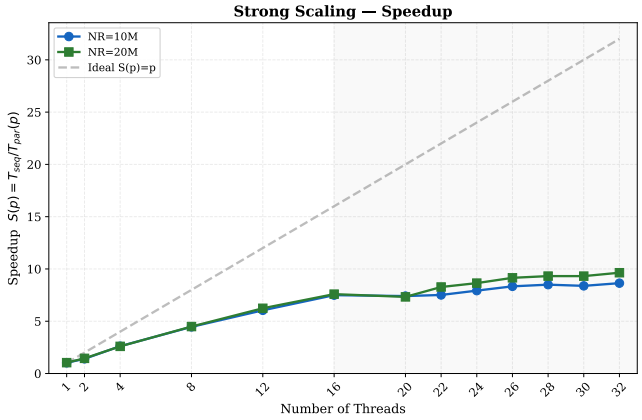
Each thread processes 1M records of R (and 2M of S). WSE $= T(1)/T(p)$; ideal is 1.0. The $p = 1$ parallel baseline (≈ 78 ms on NR=1M) incurs negligible overhead from the barrier infrastructure.

p	NR	T (ms)	WSE
1	1M	78	1.00
2	2M	106	0.74
4	4M	112	0.70
8	8M	131	0.60
10	10M	145	0.54
14	14M	155	0.51
20	20M	206	0.38
24	24M	209	0.37
32	32M	238	0.33

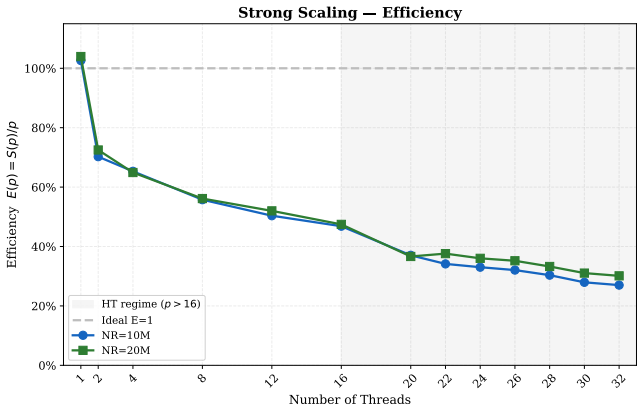
Phase Breakdown

Phase	T_{seq}	$S(14)$	$S(20)$	$S(24)$	$S(32)$
Histogram R+S	51 ms	2.0×	2.4×	3.1×	2.7×
Scatter R+S	409 ms	7.8×	9.7×	11.3×	13.7×
Join Local	309 ms	8.4×	7.2×	8.7×	8.0×
End-to-end	769 ms	6.7×	7.2×	8.7×	8.8×

Phase times from the internal timer; $T_{\text{seq}} = 769$ ms is their sum at $p = 1$; output buffers pre-allocated before timing. Merge/prefix-sum ($O(P{\cdot}k)$, a few μs) runs between phases and is not counted in any phase. Baseline differs from the strong-scaling outer timer (766 ms). Measured twice (769 and 753 ms); the higher value is used, giving slightly optimistic speedup figures (within $\leq 3\%$ run-to-run variance).



(a) Speedup $S(p) = T_{\text{seq}}/T_{\text{par}}(p)$.



(b) Efficiency $E(p) = S(p)/p$, declining from 70% at $p = 2$ to 27% at $p = 32$ (NR=10M). Shaded region: Hyper-Threading regime ($p > 16$).

Figure 1: Strong scaling — both problem sizes (NR=10M and NR=20M) follow nearly identical profiles.

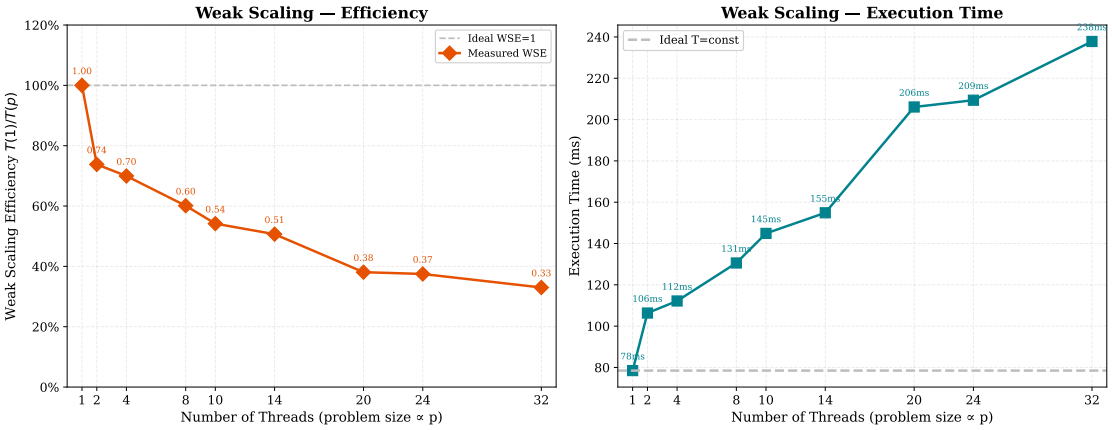


Figure 2: Weak scaling: efficiency (left) and absolute execution time (right). WSE declines monotonically; the dominant cost is scatter, whose random-write pattern saturates DRAM bandwidth as problem size grows with thread count (Section 4).

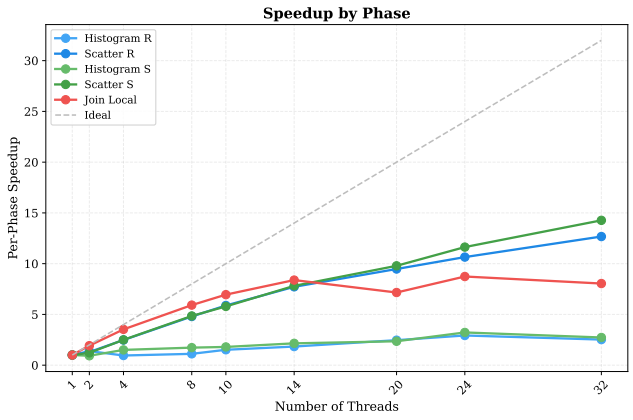


Figure 3: Per-phase speedup across $p = 1$–32. Join Local and Scatter scale well; Histogram is memory-bound.

4 Discussion

Strong Scaling and Amdahl's Law

Fitting Amdahl's law $S(p) = 1/(f + (1-f)/p)$ to the full dataset yields $f \approx 0.078$, giving $S_\infty = 1/f \approx 12.9\times$ (Figure 4). The measured peaks are 8.64× (NR=10M, $p = 32$) and 9.64× (NR=20M, $p = 32$) (Figures 1a–1b). The gap between the Amdahl prediction and the measured peak is explained by memory-bandwidth saturation: as threads increase, DRAM bandwidth is the binding constraint rather than the serial fraction.

The efficiency drops from 70% at $p = 2$ to 27% at $p = 32$ (NR=10M). With the granular thread sweep, the NUMA crossover at $p = 16 \to 20$ is directly observable: speedup reaches 7.49× (7.59× for NR=20M) at $p = 16$ (last physical-core point), then *dips* to 7.41× (7.33× for NR=20M) at $p = 20$ before recovering as HT threads improve scatter throughput. This transient regression is consistent with threads on the second socket accessing remote memory at higher latency, a hypothesis supported by the larger dip for NR=20M (larger working set, more NUMA traffic). From $p = 16$ to $p = 32$, the end-to-end time improves by only $\approx 13\%$ (102 ms → 89 ms for NR=10M; $\approx 21\%$ for NR=20M: 207 ms → 163 ms) despite doubling the logical thread count, making $p = 12$–16 (physical cores only) the practical sweet spot.

Effect of partition count ($P = 512$). At the throughput-optimal $P = 512$ (Partition Count Sensitivity subsection, below), the implementation achieves

11.4× (NR=10M) and 11.5× (NR=20M) speedup at $p = 32$, with a fitted serial fraction $f \approx 0.058$ ($S_\infty \approx 17\times$). The reduction in f relative to $P = 128$ reflects smaller per-partition `FlatCountMap` tables that fit in L3, reducing the DRAM-bound component of the join phase. At $P = 512$ the NUMA dip is entirely absent for NR=10M: speedup rises from 7.76× at $p = 16$ to 9.16× at $p = 20$, because finer partitioning keeps each `FlatCountMap` within the aggregate L3. For NR=20M the dip remains visible (9.59× → 9.22× from $p = 16$ to $p = 20$), confirming that the larger working set still generates significant NUMA traffic even at $P = 512$. Notably, the dip magnitude ($\Delta S = 0.37$) exceeds that at $P = 128$ ($\Delta S = 0.26$), indicating that finer partitioning does not reduce NUMA penalties for NR=20M.

Hyper-Threading regime (p = 20–32). The phase breakdown reveals a counter-intuitive split at $p > 16$: scatter continues to improve (9.7× → 13.7× from $p = 20$ to $p = 32$) while join peaks at $p = 14$ (8.4×), dips to 7.2× at $p = 20$ (HT onset), then fluctuates in the 7–9× range. Scatter is memory-bandwidth-bound — HT threads increase memory-level parallelism, so additional logical cores help. Join Local is compute-bound: HT pairs share execution ports, so competing hash-lookup streams contend for ALU resources rather than adding capacity. Overall, end-to-end speedup improves only moderately (7.2× → 8.8×) because the join plateau offsets much of the scatter gain.

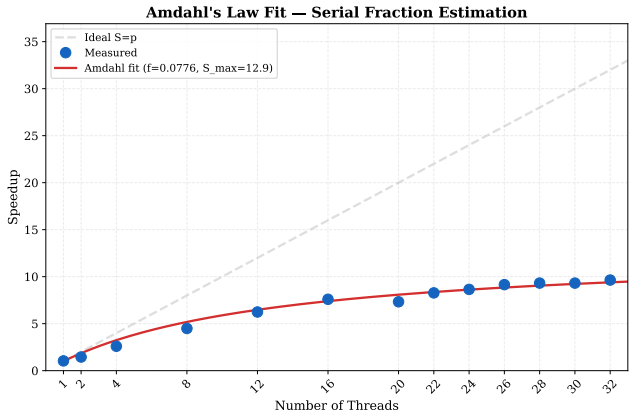


Figure 4: Amdahl's law fit (fit computed on both NR=10M and NR=20M; NR=20M curve shown, P=128). Estimated serial fraction $f \approx 0.078$; theoretical $S_\infty \approx 12.9\times$. The measured peak (9.64× at $p = 32$) falls below the prediction due to memory-bandwidth saturation.

Weak Scaling and WSE Interpretation

The $p = 1$ parallel binary takes ≈ 78 ms on NR=1M with negligible overhead from the barrier infrastructure. WSE $= T(1)/T(p)$ decreases monotonically: 0.74 at $p = 2$, 0.54 at $p = 10$, 0.33 at $p = 32$ (Figure 2).

The primary cause is scatter: as problem size grows proportionally with thread count, the random-write pattern generates increasing DRAM traffic that more cores cannot absorb (bandwidth is shared, not replicated). Histogram and join scale better at low counts because thread-local histograms and per-partition `FlatCountMap` tables (≤ 1 MB each) fit within the 20 MB shared L3 at $p \leq 4$. WSE drops most sharply between $p = 14$ and $p = 20$ (0.51 → 0.38, Δ=0.13), then levels off ($p = 20 \to 32$: Δ=0.05), coinciding with HT onset and growing NUMA penalties.

Effect of partition count on weak scaling. At the throughput-optimal $P = 512$, weak scaling improves appreciably: $T(1) \approx 90$ ms (NR=1M), WSE = 0.45 at $p = 32$ versus 0.33 at $P = 128$—a ≈ 35% improvement in scaling efficiency; in absolute execution time at $p = 32$ the gain is ≈ 15% (238 → 202 ms). The finer partitioning also produces a transient cache-benefit at $p = 4$ (WSE = 0.87 >WSE(p=2) = 0.79): more `FlatCountMap` tables fit within the aggregate L3, briefly improving efficiency before DRAM-bandwidth pressure dominates.

Phase-Level Analysis

The three computational phases (Figure 3) scale very differently, and understanding why is central to the gap between the Amdahl prediction and the measured end-to-end speedup.

Histogram is the weakest scaler (2.4–3.1× in the HT regime): one counter increment per 8-byte record makes the memory bus the bottleneck regardless of core count; at ≈ 6.6% of total time it contributes a small but measurable share of $f \approx 0.078$.

Scatter dominates the sequential baseline (≈ 53%, 409 ms) and keeps improving through the HT regime (9.7 → 13.7× from $p = 20$ to $p = 32$): the lock-free design eliminates contention entirely, and HT threads increase memory-level parallelism rather than competing for ALU resources.

The Join Local phase exhibits a contrasting scaling pattern. At $P = 128$ each per-partition `FlatCountMap` occupies ≈ 4 MB; with 20 threads the aggregate (80 MB) far exceeds the 20 MB shared L3. The phase scales near-linearly to $p = 14$ (8.4×) then plateaus: HT pairs share execution ports, adding no compute capacity. At $P = 512$ each table shrinks to ≈ 1 MB and the aggregate fits in L3, explaining the performance peak (Partition Count Sensitivity subsection, below).

Partition Count Sensitivity

With $P \in [16, 4096]$ (at $p = 20$, NR=10M), execution time reaches a minimum at $P = 512$–1024 (≈ 86 ms), with a variation of < 1.6× across the full range. At $P = 16$ each partition holds ≈ 625K records and the `FlatCountMap` occupies ≈ 32 MB, spilling into DRAM. The optimum at $P = 512$ corresponds to ≈ 20K records/partition, with the hash table (≈ 1 MB) fitting well within the shared 20 MB L3. Beyond $P = 1024$ the overhead of repeated `FlatCountMap` allocation/deallocation cycles grows: at $P = 4096$ each thread performs $\lceil P/p \rceil = \lceil 4096/20 \rceil = 205$ constructor/destructor calls per join phase (vs. $\lceil 512/20 \rceil = 26$ at $P = 512$), increasing TLB pressure and loop overhead, raising time to 89 ms at $P = 2048$ and 94 ms at $P = 4096$.

A non-monotonic bump at $P = 32$ (130 ms, slower than $P = 16$) arises from a two-effect transition. At $P = 16 < p$=20: the cyclic assignment exhausts all partitions after 16 threads, leaving 4 threads idle throughout the join phase; the 16 active threads each execute one build-probe cycle on a ≈ 32 MB `FlatCountMap` (single allocation, DRAM-bound). At $P = 32$ ($\gtrsim p$=20): all 20 threads are active but 12 of them process two partitions sequentially, each requiring a fresh 16 MB `FlatCountMap` allocation; the maximum thread workload (≈ 625 K records) is nearly identical to P=16, yet the extra malloc–free cycle between partitions adds construction overhead without any cache benefit, since 16 MB tables are still firmly DRAM-bound. From $P = 64$ onwards, per-partition tables shrink below ≈ 8 MB and begin to overlap with the shared 20 MB L3, so additional allocation cycles are more than compensated by reduced DRAM traffic, restoring monotone improvement. The minimum lies in $P \in [512, 1024]$ (≈ 86 ms); timing improves by ≈ 20% from $P = 128$ (107 ms) to $P = 512$, so the choice of P is meaningful in the sub-optimal range (Figure 5).

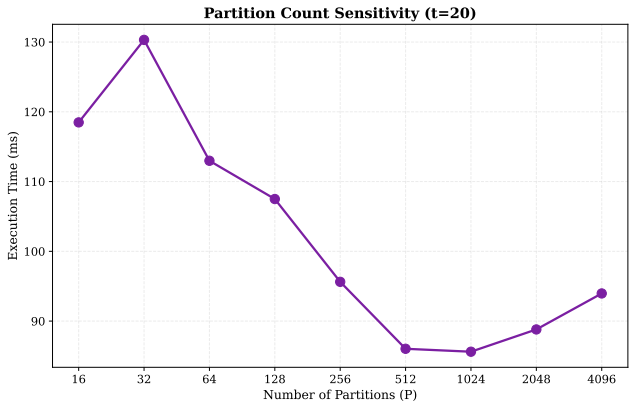


Figure 5: Partition count sensitivity ($p = 20$, NR=10M). Minimum at $P = 512$–1024; non-monotonic bump at $P = 32$ reflects a cache-occupancy transition.

Duplicate Density

Sequential time is essentially constant for `max_key` ≤ 100K (≈ 697–721 ms): with many duplicates, few distinct keys keep the per-partition `FlatCountMap` sparsely occupied and most probes terminate quickly. Speedup peaks at `max_key`=10K (7.5×), where key values distribute uniformly across $P = 128$ partitions and per-partition tables are neither too sparse nor too densely populated. At `max_key`=10M (near-unique keys) the hash table is fully populated, increasing sequential time by +29% (≈ 708 ms → 915 ms, relative to the peak-speedup point at `max_key`=10K) while parallel execution time rises by +42% (≈ 94 ms → 133 ms)—a larger relative increase—because at $P = 128$ each per-partition `FlatCountMap` occupies ≈ 4 MB; with 20 active threads the aggregate working set (20 × 4 MB=80 MB) far exceeds the 20 MB shared L3, so the working set remains DRAM-bound at any key density and the parallel version cannot exploit sparser table occupancy. As a result, speedup at `max_key`=10M settles at ≈ 6.9×, modestly below the 7.5× peak, showing that the partitioned approach works well regardless of key distribution. At the optimal $P = 512$, per-partition tables shrink to ≈ 1 MB and can partially reside in L3 at near-unique densities, explaining the substantially higher speedup observed in that regime (Figure 6).

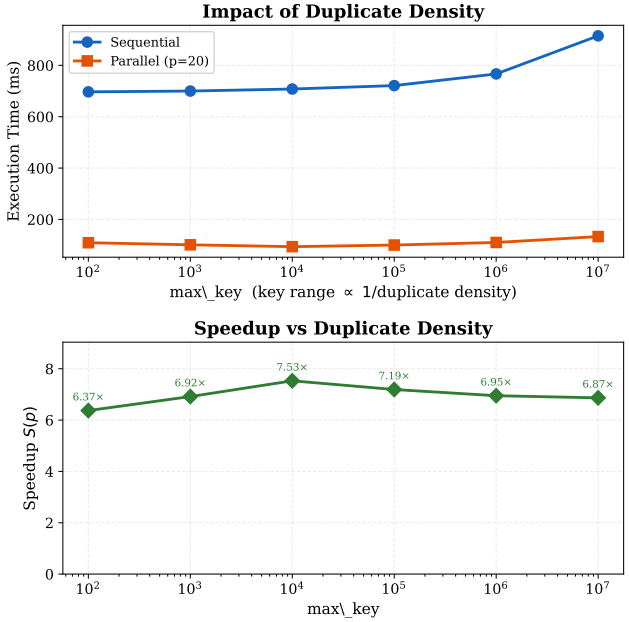


Figure 6: Impact of duplicate density ($p = 20$, NR=10M, $P = 128$). Top: absolute execution time (seq and par). Bottom: speedup vs. `max_key`. Speedup peaks at `max_key`=10K (7.5×); drops to 6.4× at `max_key`=100 due to partition load imbalance (100 distinct keys for 128 partitions); settles at 6.9× at `max_key`=10M (near-unique) because per-partition tables remain DRAM-bound at $P = 128$ regardless of key density.